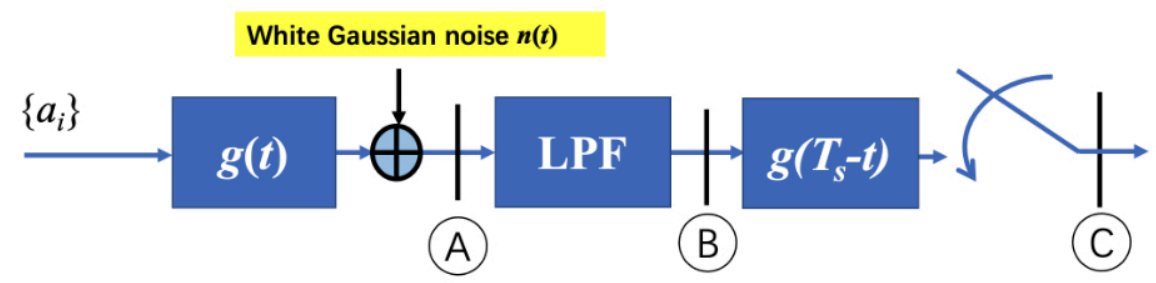


1. In this problem, we investigate the SNR (Signal-to-Noise Ratios) at different points in a baseband transmission system with the matched filter and assume no-ISI.

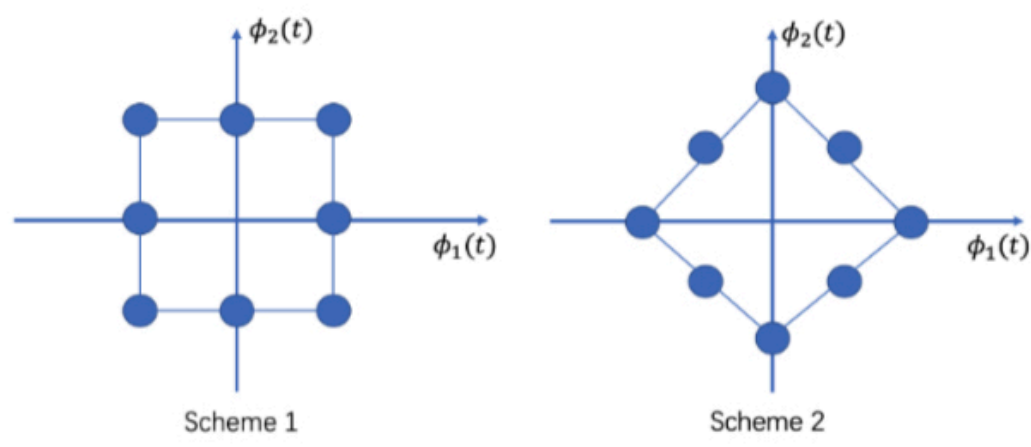


The baseband waveform is a square-root raised cosine with roll-off factor $\alpha \in [0, 1]$. At the receiver, the signal is sampled at T_s after the matched filter. The symbols $\{a_i\}$ take values in $\{0, 1, 2, 3\}$ with the equal probability. The double-sided PSD (Power Spectrum Density) of the noise is $n_0/2$.

- With the no-ISI condition, what is the bandwidth W of $G(f)$? Write the explicit form of the frequency representation $G(f)$.
- Derive the SNR at point B, from the baseband waveform perspective. You can keep the integral term of $G(f)$. (Hint: The distribution of $\{a_i\}$ should be considered.)
- Derive the SNR at point C, after sampling at T_s . Compare the SNR at point A, B, C and explain.

$$\begin{aligned}
 a) \quad & \sum_{n=-\infty}^{\infty} G\left(f - \frac{n}{T_s}\right) = T_s, \quad W = \frac{1+\alpha}{2T_s} \\
 b) \quad & S = \frac{E\{|a_i|^2\}}{T_s} \int_{-\infty}^{\infty} |G(f)|^2 df, \quad N = \frac{n_0}{2} \cdot 2W = n_0 W \\
 SNR_B &= \frac{E\{|a_i|^2\} \int_{-\infty}^{\infty} |G(f)|^2 df}{n_0 W T_s} = \frac{\frac{1}{2} \int_{-\infty}^{\infty} |G(f)|^2 df}{n_0 (1+\alpha)} = \frac{1}{n_0 (1+\alpha)} \int_{-\infty}^{\infty} |G(f)|^2 df \\
 c) \quad & S = E\{|a_i g_o(t_0)|^2\} = E\{|a_i|^2\} \left| \int_{-\infty}^{\infty} g(x) dx \right|^2 = E\{|a_i|^2\} \left(\int_{-\infty}^{\infty} |G(f)|^2 df \right)^2 \\
 N &= \frac{n_0}{2} \int_{-\infty}^{\infty} |G(f)|^2 df \\
 SNR_C &= \frac{2E\{|a_i|^2\} \int_{-\infty}^{\infty} |G(f)|^2 df}{n_0} = \frac{1}{n_0} \int_{-\infty}^{\infty} |G(f)|^2 df \\
 SNR_A &= \frac{E\{|a_i|^2\} \int_{-\infty}^{\infty} |G(f)|^2 df}{T_s \cdot \frac{n_0}{2} \int_{-\infty}^{\infty} |G(f)|^2 df} = 0 \quad (\text{From the baseband waveform})
 \end{aligned}$$

4. Below are two schemes of 8-symbol constellation. Explain which one is better, in terms of SEP (Symbol Error Probability), given the same power constraint.



Scheme 1 can be rotated 45° to get Scheme 2

Therefore, $SEP_1 = SEP_2$

2. In a baseband transmission system, the transmitted signal is

$$x = \begin{cases} 0, & \text{symbol '0' is transmitted} \\ A, & \text{symbol '1' is transmitted} \end{cases}$$

and the received signal is $y = x + n$, where n is the equivalent additive noise, following the Laplace distribution

$$f(n) = \frac{1}{2\lambda} e^{-\frac{|n|}{\lambda}}$$

- Assume that the transmitted symbols are equiprobable. Determine the detection threshold with minimum SEP (Symbol Error Probability) and calculate the corresponding SEP.
- If the detection rule at the receiver is biased, specifically

$$\tilde{x} = \begin{cases} '0', & y < \frac{1}{2}A \\ '1', & y \geq \frac{1}{2}A \end{cases}$$

- Find the conditional symbol error probability $P(\tilde{x} = '0'|x = '1')$ and $P(\tilde{x} = '1'|x = '0')$.
- Find the prior distribution of the transmitted symbols so that the detection rule in b) minimizes the SEP for the given prior distribution. (Hint: the question is different from minimizing the SEP given the detection rule)

$$a) \quad P(y|\pi=0) P(x=0) \underset{A}{\overset{0}{>}} P(y|\pi=A) P(\pi=A)$$

$$\Leftrightarrow P(y|\pi=0) \underset{A}{\overset{0}{>}} P(y|\pi=A)$$

$$\Leftrightarrow f(y) \underset{A}{\overset{0}{>}} f(y-A) \Leftrightarrow \frac{1}{2\lambda} e^{-\frac{|y|}{\lambda}} \underset{A}{\overset{0}{>}} \frac{1}{2\lambda} e^{-\frac{|y-A|}{\lambda}} \Leftrightarrow |y-0| \underset{A}{\overset{0}{>}} |y-A| \Leftrightarrow y \underset{A}{\overset{0}{>}} \frac{A}{2} \Rightarrow \hat{x} = \begin{cases} '0', & y < \frac{A}{2} \\ '1', & y > \frac{A}{2} \end{cases}$$

$$\text{detection threshold} = \frac{A}{2}$$

$$SEP = P(\pi=A \& y < \frac{A}{2}) + P(\pi=0 \& y > \frac{A}{2})$$

$$= P(n < -\frac{A}{2} | \pi=A) P(\pi=A) + P(n > \frac{A}{2} | \pi=0) P(\pi=0)$$

$$= \frac{1}{2} (1 - P(-\frac{A}{2} \leq n \leq \frac{A}{2}))$$

$$= \frac{1}{2} \left(1 - \int_{-\frac{A}{2}}^{\frac{A}{2}} \frac{1}{2\lambda} e^{-\frac{|n|}{\lambda}} dn \right) = \frac{1}{2} \left(1 - \int_0^{\frac{A}{2}} \frac{1}{\lambda} e^{-\frac{n}{\lambda}} dn \right) = \frac{1}{2} e^{-\frac{A}{2\lambda}}$$

$$\begin{aligned}
 b) \quad & P(\hat{x}='0' | \pi='1') = P(y < \frac{A}{2} | \pi=A) = P(n < -\frac{3}{2}A) = \int_{-\infty}^{-\frac{3}{2}A} \frac{1}{2\lambda} e^{-\frac{|n|}{\lambda}} d\lambda = \frac{1}{2} e^{-\frac{3A}{2\lambda}} \\
 & P(\hat{x}='1' | \pi='0') = P(y \geq \frac{A}{2} | \pi=0) = P(n \geq \frac{A}{2}) = \int_{\frac{A}{2}}^{\infty} \frac{1}{2\lambda} e^{-\frac{|n|}{\lambda}} d\lambda = \frac{1}{2} e^{-\frac{A}{2\lambda}}
 \end{aligned}$$

$$c) \quad P(\tilde{x}='0' | \pi=0) P(\pi=0) \underset{1}{\overset{0}{>}} P(\tilde{x}='1' | \pi=A) P(\pi=A)$$

$$\Leftrightarrow f\left(\frac{A}{2}\right) P(\pi=0) = f\left(-\frac{3}{2}A\right) P(\pi=A)$$

$$\Leftrightarrow \frac{P(\pi=0)}{P(\pi=A)} = e^{-\frac{A}{2\lambda}}$$

$$\Leftrightarrow P(\pi=0) = \frac{e^{-\frac{A}{2\lambda}}}{1 + e^{-\frac{A}{2\lambda}}}, \quad P(\pi=A) = \frac{1}{1 + e^{-\frac{A}{2\lambda}}}$$

5. In this problem, we compare different methods in passband modulation. Assume the double-sided PSD of noise is $n_0/2$ and all the symbols are equiprobable.

- Given the energy per symbol E_s , find the SEP (Symbol Error Probability) of 4QAM and 16QAM. Explain why the SEP is lower for 4QAM.
- Given the energy per bit E_b , find the SEP of 4QAM and 16QAM. Which one has a lower SEP? (Hint: it is recommended to draw the SEP-SNR curves using Matlab)
- Given the energy per bit E_b , compare the SEP of 16QAM and 16PSK. Explain intuitively why QAM is better?

$$a) \quad P_{S, MQAM} = 4 \left(1 - \frac{1}{\sqrt{M}} \right) Q \left(\sqrt{\frac{3E_s}{(M-1)n_0}} \right)$$

$$\left. \begin{aligned} P_{S, 4QAM} &= 2 Q \left(\sqrt{\frac{E_s}{n_0}} \right) \\ P_{S, 16QAM} &= 3 Q \left(\sqrt{\frac{E_s}{5n_0}} \right) \end{aligned} \right\} \Rightarrow P_{S, 4QAM} < P_{S, 16QAM}$$

$$b) \quad P_{S, 4QAM} = 2 Q \left(\sqrt{\frac{2E_b}{n_0}} \right)$$

$$P_{S, 16QAM} = 3 Q \left(\sqrt{\frac{4E_b}{5n_0}} \right)$$

$$P_{S, 4QAM} < P_{S, 16QAM}$$

$$c) \quad P_{S, MPSK} = 2 Q \left(\sqrt{\frac{2E_b}{n_0}} \sin \frac{\pi}{M} \right)$$

$$P_{S, 16PSK} = 2 Q \left(\sqrt{\frac{PE_b}{n_0}} \sin \frac{\pi}{16} \right) > 3 Q \left(\sqrt{\frac{4E_b}{n_0}} \right) = P_{S, 4QAM}$$

3. In a baseband transmission system with binary antipodal symbol set, the data rate is 5kbps and

the double-sided PSD (Power Spectrum Density) of the additive Gaussian noise is $n_0/2$, where $n_0 = 10^{-9} \text{ W/Hz}$.

- If the system is required to be reliable, producing less than 100 bit errors on average in one day, what is the required SEP (Symbol Error Probability) of the system?
- Determine the minimum energy per symbol E_s and the average transmitting power P_t . (Hint: use `qfuncinv` in Matlab to calculate)

$$a) \quad SEP = Q \left(\sqrt{\frac{2E_s}{n_0}} \right) = Q \left(\sqrt{\frac{2E_b}{n_0}} \right) = P_b \leq \frac{100 \text{ bits}}{5 \text{ kbps} \times 86400 \text{ s}} = 2.31 \times 10^{-7}$$

$$b) \quad E_{s, \min} = \frac{n_0}{2} \left[Q_{\text{funcinv}}(SEP) \right]^2 = 1.27 \times 10^{-8} \text{ J}$$

$$P_t = E_s R_s = 6.4 \times 10^{-3} \text{ W}$$